

A Computer-Assisted Proof of a First-Player Win in Infinite Freestyle Gomoku

Gaoqiang Liu
University of Bayreuth, Bayreuth, Germany
Gaoqiang.Liu@uni-bayreuth.de

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Abstract

We present a computer-assisted proof that the first player wins freestyle Gomoku on the infinite integer lattice. Players alternately occupy an empty lattice point, and a consecutive run of at least five stones in any of four directions wins immediately; neither player has forbidden moves. The certified strategy starts at the origin and wins before the second player within 35 placements in total. The finite proof object uses a conservative abstraction which truncates adjacent coordinate gaps to five, separately on the two axes. We prove preservation of winning lines, coverage of every legal second-player placement, and implementability of relative first-player actions in every concrete preimage. Thus distant play is represented rather than treated as a harmless pass. Symmetry reduces the first reply to 20 orbits covering 120 normalized positions. Bounded threat rules compress parts of the strategy while retaining counterthreats and actual move counts. A checker independent of the search reconstructs successor sets and checks the complete dependency closure. We supply the mathematical soundness argument, the recorded verification, and the complete certificate with frozen verification code. The bound is an upper bound, not an assertion of optimality.

1 Introduction

Gomoku is a strong positional game: either player can win by completing a prescribed line, and the order in which the two players complete their lines matters. This paper concerns the infinite board, not an extension of the boundary rules of a finite board. Both players can use every point of $\mathbb{Z}^2$. In particular, the second player can build a threat outside the region in which the first player’s attack began, force defensive moves, and later connect stones that were previously separate. A proof must account for all these possibilities and for a second-player win that occurs first.

The finite-board result is an important computational precedent. Allis, van den Herik, and Huntjens solved two unrestricted variants on a 15×15 board, including the variant in which overlines win [1, 2]. Their threat-space search and proof-number search also suggest how to generate strategy candidates. A finite-board strategy, however, is not by itself an infinite-board strategy. Additional cells create additional winning lines for the second player as well as additional legal replies. A boundary may assist the first player’s defensive obligations; removing it is not a monotone change in the strong game.

Our contribution is a finite, independently checkable proof object for the infinite freestyle rules. Its central mathematical device is a dynamic coordinate-gap abstraction. It retains every stone of both colors, records all small coordinate differences exactly, and uses finite conservative transition sets for large gaps. A first-player action is specified relative to an existing stone and is checked

against every possible abstract outcome. The strategy therefore has an interpretation on every actual lattice position represented by a certificate node. The abstraction may include additional outcomes, but it must cover every actual outcome and certify a continuation for each listed one.

The main result is the following.

Theorem 1.1 (Infinite freestyle Gomoku). *On $\mathbb{Z}^2$, with Black moving first, alternating placements on empty points, no forbidden moves, and a consecutive run of at least five stones in one of the horizontal, vertical, or two diagonal directions winning immediately, Black has a strategy with first move $(0,0)$ which wins before White against every legal White strategy. The win occurs after at most 35 actual placements by the two players combined, counted from the empty board.*

The proof has two parts. The mathematical part proves that the accepted certificate rules imply a strategy on the actual infinite board. The computational part verifies a particular finite certificate satisfying those rules. The checker does not trust engine evaluations, search success labels, or omitted branches. The conclusion does not assert that 35 is the smallest possible uniform bound, nor that every play lasts 35 moves.

1.1 Related work and the scope of the claim

The literature review distinguishes strong play, where either player may win first, from weak Maker–Breaker games, in which the second player’s only role is to obstruct the first player. Results for the weak game do not automatically settle the strong game. Erde’s work on a biased n -in-a-row game also uses a different move protocol [3]. Pairing strategies for larger targets [5] concern defensive strategies and a different target length. Surveys and accounts dated 2014, 2019, and 2022 describe the infinite strong five-in-a-row problem as open [4, 5, 6]. These references document the status reported at those dates.

An additional logical distinction concerns bounded play. A winning strategy in a general infinitely branching game need not have a uniform finite winning-time bound merely because each individual play terminates. The locality result discussed by Hamkins and Leonessi explicitly excludes Gomoku in its treatment of simple stone-placing games [7]. Our uniform bound comes from the actual ranks and bounded strategies of the certificate, not from a compactness assertion about arbitrary Gomoku strategies.

In particular, the strong-to-weak transfer removes the obligation to stop the opponent’s own line. It does not justify the reverse transfer. The finite 15×15 solution therefore supplies an infinite-board weak-game precedent, while the present strong-game proof must add the White-first-win checks. Allis and coauthors describe a 35-ply principal line and report solution-tree depth 35 for both finite-board variants in Table 5 [1]. Their tree uses threat-space compression and reduced White-move selection, including treatment of defensive counterfours. These finite-board statistics do not establish an infinite-board bound. Our 35 counts every actual placement, including counterfour interruptions and the final winning move, against every infinite-board defense.

2 Rules and finite positions

Let

$$D = \{(1, 0), (0, 1), (1, 1), (1, -1)\}.$$

A five-segment is a set $L = \{s + id : 0 \leq i \leq 4\}$, where $s \in \mathbb{Z}^2$ and $d \in D$. A finite position is a pair $P = (B, W)$ of finite, disjoint sets of points occupied by Black and White. Write $|P| = |B| + |W| = p$ for its actual placement count. Reachable nonterminal positions have either $|B| = |W|$, with Black

to move, or $|B| = |W| + 1$, with White to move. A color has won if its set contains a five-segment. This definition includes all longer consecutive runs because each contains a five-segment. Every play ends at its first winning placement.

For finite disjoint occupied sets $S, T \subseteq \mathbb{Z}^2$, with S denoting the side under consideration and T its opponent, assume S contains no five-segment and define

$$\text{Win}(S, T) = \{q \in \mathbb{Z}^2 \setminus (S \cup T) : S \cup \{q\} \text{ contains a five-segment}\}.$$

This is the set of immediate winning placements. It is finite for finite S under the stated nonwinning hypothesis: each such placement completes a five-segment with four old stones. The notation is also used for virtual finite positions introduced by threat rules; virtual stones are always identified as such and do not change the count p of actual placements. Those uses satisfy the same nonwinning hypothesis. An already winning set is terminal and is tested directly, not passed to this definition.

The restriction on Black actions in the certificate language is a restriction on the strategy being certified, not on the game. Black is legally allowed to play anywhere. It suffices to certify one winning strategy using a smaller action language. White is subject to no analogous search restriction: every legal White point must be covered.

3 Dynamic coordinate gaps

3.1 Normalization and concrete preimages

For either coordinate axis, sort the distinct coordinate values of all points of $B \cup W$, together with the constant anchor 0:

$$v_0 < v_1 < \dots < v_m.$$

If $v_j = 0$, define a strictly increasing map f on these values by

$$f(v_j) = 0, \quad f(v_{i+1}) - f(v_i) = \min(v_{i+1} - v_i, 5).$$

Applying the two maps independently gives the normalization

$$\alpha(P) = (\{(f_x(x), f_y(y)) : (x, y) \in B\}, \{(f_x(x), f_y(y)) : (x, y) \in W\}).$$

The anchor is a coordinate convention, not an extra stone. Normalization retains colors, stone identities, occupancy, turn, and actual placement count. A normalized position Q represents every P with $\alpha(P) = Q$. A normalized adjacent gap equal to five represents any actual adjacent gap of length at least five. Thus an abstract position generally has infinitely many concrete preimages.

Lemma 3.1 (Exact short coordinate differences). *For any two old coordinate values u, v , if $|u - v| \leq 4$, then $f(v) - f(u) = v - u$. Conversely, if $|f(v) - f(u)| \leq 4$, then the same equality holds. If $|u - v| > 4$, then $|f(v) - f(u)| > 4$.*

Proof. Assume $u < v$ and sum the adjacent gaps between them. If their sum is at most four, every gap is less than five and none is changed. If a gap of length at least five occurs, its normalized length alone is five, so the normalized sum exceeds four. If no such gap occurs, the sum is unchanged. This proves all assertions; the reversed ordering follows by changing sign. $\square$

Lemma 3.2 (Winning-line and occupancy preservation). *Normalization is injective on old points and preserves, in both directions, the existence of a five-segment of either color.*

Proof. Each axis map is strictly increasing, so two different old points remain different. In a five-segment, every pair has absolute difference at most four on each axis. Lemma 3.1 preserves these signed differences, including zero differences. Hence the image is a five-segment of the same direction. Conversely, the signed differences among any five points which form a normalized five-segment are all at most four, so their actual differences are exactly those of that segment. The same argument applies to each color separately. A run longer than five contains a five-segment, so overlines require no special exception. $\square$

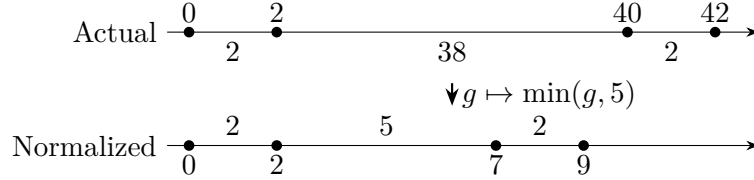


Figure 1: One-axis normalization. The upper drawing is schematic, not to scale. Small gaps remain exact; the gap of length 38 becomes a saturated gap of length five. The operation is applied to the coordinate values of both colors and the anchor, not just to the attacking stones.

3.2 All White placements have finite representatives

Suppose the old coordinate values are normalized. An inserted coordinate is equal to an old value, lies outside both extremes, or splits one old adjacent gap. Equal-value insertions are retained because a point with the same x or y coordinate need not be occupied. Beyond an extreme, the distance is represented by one of 1, 2, 3, 4, 5. An exact gap $g < 5$ is split at each integer u with $1 \leq u < g$. A saturated gap is split into truncated parts (a, b) with

$$1 \leq a, b \leq 5, \quad a + b \geq 5. \quad (1)$$

For each insertion type, the old values and the new value are renormalized together, keeping the anchor at zero. Take the Cartesian product of the two axis insertion sets, transform all old stones, add the new White stone, and discard occupied targets. Duplicate normalized child states are merged. Denote the resulting finite set by $\text{Succ}_W(Q)$.

Lemma 3.3 (Complete White coverage). *For every preimage P of Q and every legal actual White point q , $\alpha(B, W \cup \{q\}) \in \text{Succ}_W(Q)$.*

Proof. On each axis classify the actual new coordinate as above. Equal coordinates are included. An exterior distance u has type $\min(u, 5)$. In an exact old gap, its length and every possible integer split are unchanged. In an actual saturated gap $G \geq 5$, write the new positive lengths as $u + v = G$. Their truncations $a = \min(u, 5)$ and $b = \min(v, 5)$ satisfy (1). The unsplit gaps are unchanged in type, so this insertion gives exactly the normalized old coordinates and new coordinate obtained from the actual position. The two axes can be treated independently. Injectivity ensures that a legal actual target is not removed by the occupancy filter. $\square$

The converse for a fixed preimage is unnecessary: the finite successor set may contain extra children. Those children impose extra proof obligations. They never justify removal of an actual reply. The independent checker constructs its axis types differently: it realizes an old saturated gap at every concrete length from five through ten and renormalizes each integer insertion. This

produces the same types as (1). Indeed each such pair has a realization of length $a + b \in [5, 10]$, and any larger gap has one of the same truncated pairs. Unsplit saturated gaps can be realized at five without changing these local insertion data. This finite enumeration is accompanied by the argument above; checking a handful of large example distances alone would not prove coverage of all distances.

3.3 Implementable Black actions and their uncertainty

A relative Black action is a pair (r, δ) , where r is an old occupied point in the normalized position and $\delta \in \mathbb{Z}^2$ satisfies $\|\delta\|_\infty \leq 4$. It designates $r + \delta$, which must be empty in the normalized position. In a concrete preimage, Black uses the actual stone corresponding to r and places at $r_{\text{act}} + \delta$. Both colors may provide a reference stone. Only the first move on the empty board has the special absolute instruction $(0, 0)$.

Lemma 3.4 (Relative-action legality and coverage). *The designated Black placement is legal in every preimage. Filtering the complete two-axis insertion types by $q'_i - f'_i(r_i) = \delta_i$ for each axis i gives a finite set $\text{Succ}_B(Q, r, \delta)$ which contains every actual normalized outcome of that placement. All outcomes in this set must be certified.*

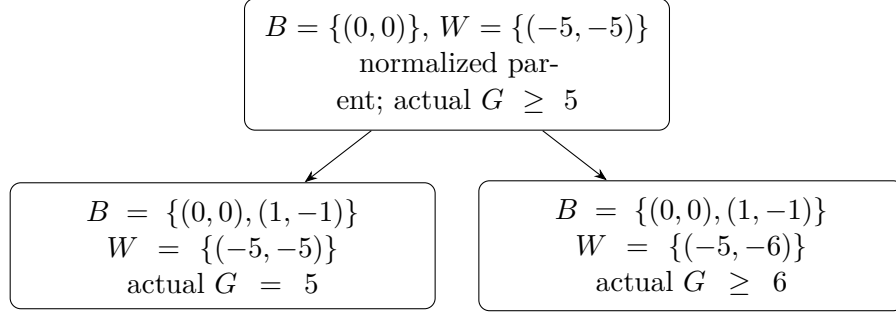
Proof. An old actual stone occupying $r_{\text{act}} + \delta$ has signed differences δ_i from the reference, each of absolute value at most four. Lemma 3.1 would place its image at $r + \delta$, contradicting the checked empty target. Conversely an old image at that target would give exactly that actual displacement. The action is therefore legal uniformly. After inserting the actual target, the target–reference differences are still small and exact, so its axis insertion types pass the filter. Lemma 3.3’s insertion argument is independent of the color of the added stone and proves coverage. Filtering can retain spurious outcomes, which are handled by requiring every retained child. $\square$

An action can have more than one normalized outcome even though the actual strategy makes a single definite placement. For example, let Black have $(0, 0)$ and let White have $(-60, -G)$, with $G \geq 5$. Every such position normalizes to Black at $(0, 0)$ and White at $(-5, -5)$. Place Black at $(1, -1)$ relative to the origin. If $G = 5$, White normalizes to $(-5, -5)$ afterwards; if $G \geq 6$, White normalizes to $(-5, -6)$. The gap that was saturated has been split by the new coordinate. A strategy certificate must cover both outcomes. A search that keeps only the first displayed outcome would be unsound.

3.4 Finiteness and the role of distant play

For an ordinary actual position with p stones, either axis has at most $p + 1$ old coordinate values, including the anchor. Every adjacent normalized gap is at most five, so each normalized coordinate lies in $[-5p, 5p]$. Thus there are finitely many normalized colored positions for $p \leq 35$, although their number is large. This count concerns ordinary actual positions; the virtual White unions inside a threat witness do not have cardinality p and are handled by their separate finite closures. The certificate selects only a finite subset. Every explicit edge adds exactly one actual stone, and its placement-count rank strictly increases.

This abstraction retains distant White stones. When further White moves split their gaps, the resulting children remain among the complete insertion types. A distant three, a boundary-crossing five, or the later connection of separate White groups is consequently tested through actual geometric successor obligations. There is no fixed boundary and no permission to identify all exterior play with a pass. Remote compression rules used later require separate sufficient conditions and explicitly account for defensive interruptions.



Black's actual action is $(r, \delta) = ((0, 0), (1, -1))$.
Both normalized children require a proof.

Figure 2: A definite actual Black placement can have several abstract outcomes. This is a geometric example of the certificate's universal outcome obligation, not a claim that White stones are moved on the actual board.

4 Certificate language and soundness

4.1 Explicit nodes and modular files

An ordinary certificate is a finite directed graph whose nodes record a normalized colored position, the player to move, and a rule kind. Its header specifies the freestyle rules and an integer total-placement bound H . We use $H = 35$. A nonterminal explicit Black node stores a legal relative action and an edge to every state of $\text{Succ}_B(Q, r, \delta)$. A nonterminal explicit White node stores an edge to every state of $\text{Succ}_W(Q)$. Duplicate child positions are rejected, and the set of recorded child positions must equal the set reconstructed by the checker. Every edge increases p by one. The checker rejects positions with White already winning, and rejects continued play after a Black win. Terminal nodes must have a Black five, the correct turn parity, and no White five; they are not accepted merely because a label says “terminal”.

To keep individual files manageable, a bundle records one such Black or White node and points to complete child certificates. Each reference contains the child position, path, and SHA-256 digest of the actual child bytes. A child must have the stated root and a bound no greater than its parent's. Missing files, mismatching digests, mismatching roots, incomplete geometric coverage, and cyclic file references are rejected. Sharing a completed subproof is permitted. Sharing an unproved search state is not. Hashes bind data; they do not themselves prove the mathematics.

4.2 Bounded rules and actual clocks

Some accepted node kinds stand for a finite winning continuation without explicitly listing every short reply. For example, after Black has two distinct immediate winning points, White cannot occupy both in one move; provided White has no immediate winning placement, Black wins within two further actual placements. Other rules retain the necessary counterthreat and blocker replies explicitly. Sections 5 and 6 state the sufficient conditions.

A bounded rule at actual count p must provide a strategy valid for every actual preimage and every subsequent legal White move. It must rule out White winning first at every intermediate turn and bound the number of further actual placements by h , with $p + h \leq H$. The count h includes both colors, defensive interruptions, and the move on which Black wins. A virtual union of possible White defense points is only a conservative simulation state. Adding three virtual points

to that union does not mean that White made three actual moves, nor that White made zero moves. Termination is established using the actual strategy clock.

Theorem 4.1 (Certificate soundness). *Assume every bounded rule used by a finite accepted certificate has the uniform preimage soundness property just described. A certificate rooted at normalized Q with total bound H supplies an actual Black winning strategy from every nonterminal preimage P of Q , in at most $H - |P|$ further actual placements. Black wins before White.*

Proof. Use induction on $H - p$. At an accepted Black terminal, winning-line preservation gives a Black five in the actual position and excludes a White five. A bounded-rule node supplies the claimed actual strategy directly, within its checked clock. At an explicit Black node, the relative action is legal in every preimage by Lemma 3.4. Its actual normalized result is one of the certified children; apply induction with one smaller remaining count. At an explicit White node, whatever legal point White chooses, Lemma 3.3 puts the result among the children. Each certified child excludes a White win by Lemma 3.2 and has smaller remaining count. Symbolic White rules partition replies into explicitly covered children and uniformly bounded winning continuations, so the same argument applies. A symmetry frame, when used by a rule, transports all actual placements by its checked lattice isometry and transports them back without changing the clock. Finite ranks preclude an infinite explicit path. Bundle references are the same induction across file boundaries, with roots and bounds checked and file cycles excluded. Therefore all actual plays terminate in a Black win within the stated bound. $\square$

The soundness theorem separates the universal mathematical statement from the finite computation. It is not a statement that a heuristic search which returns success must be correct. Only a fully accepted certificate can instantiate the theorem. UNKNOWN, timeout, untried actions, incomplete branches, and candidate proofs cannot instantiate it.

5 Short threat rules

In this section $\text{Win}(X, Y)$ is used only when X has no five. The normalization map is α ; the exhaustive one-move lifting result established earlier routes actual replies to normalized children.

For a full position after $p = |B| + |W|$ actual moves with total limit H , every rule that omits a continuation of at most c moves must satisfy $p + c \leq H$. The short rules below use $c = 2$ or $c = 4$ as stated.

Lemma 5.1 (rigid old components). *Form the graph on all old occupied points, including White connector stones, with an edge whenever the Chebyshev distance is at most 4. Each connected component has the same relative coordinates in every integer preimage, and hence moves as a rigid frame under its own translation. No gap displayed as 5 lies strictly within the axis projection of a component. Distinct components can have different translations and must not be treated as one rigid frame.*

Proof. By Lemma 3.1, both signed coordinate differences of every edge are exact in every preimage. Summing these displacement vectors along a connected path fixes the difference between any two vertices of its component. If a gap displayed as 5 lay inside a component's axis projection, a path between vertices on opposite sides would have an edge crossing that gap. That edge would have axis difference at least 5, contrary to its Chebyshev distance being at most 4. $\square$

Lemma 5.2 (Lifting immediate completions). *Let $Q = \alpha(P)$ be nonterminal. For either color, let S_P, T_P and S_Q, T_Q denote its own and opposing occupied sets in P and Q . Every completion*

$q \in \text{Win}(S_Q, T_Q)$ has four old supporting stones in one rigid component. Choose a supporter r adjacent to q and lift q as $r_{\text{act}} + (q - r)$. The result is independent of the witness and reference, is empty, and completes five in every preimage. For each fixed preimage this gives a bijection with $\text{Win}(S_P, T_P)$; in particular, distinct completions stay distinct even when supported in different components.

Proof. The four supporters lie in one five-segment. Their signed differences are exact by Lemma 3.1, so the prescribed hole offset completes their translated segment. If the lifted hole were occupied, its old occupant would be within distance one of r and would occupy the displayed hole, a contradiction. Two witnesses for the same displayed hole have adjacent supporters within distance two of one another, so they belong to the same rigid component and give the same lift. Within one component the lift is a translation. If holes from different components collided after lifting, their adjacent supporters would have actual distance at most two; the short-distance lemma would join those components, again a contradiction. Conversely, the four supporters of any actual completion have exact internal differences. Its hole relative to an adjacent supporter is empty in Q (an occupying old stone would also lift to the actual hole), and completes the displayed segment. Thus every actual completion is included. $\square$

All propositions in this section concern a nonterminal position with White to move. Every retained critical move is sent to every abstract successor of its anchored action. Soundness of this routing uses the exhaustive one-move dynamic-gap transition lemma; the geometric arguments below explain why replies outside the listed actions can be omitted. A chosen action is an old occupied reference point plus an offset of Chebyshev norm at most 4. Different preimages may produce different normalized children, and all those children are required.

5.1 Immediate completion rules

Proposition 5.3 (double completion). *If White has no immediate winning move and Black has at least two distinct immediate winning points, Black wins within the next two actual moves.*

Proof. White cannot win on its present move. It occupies at most one of Black's two points. Black occupies another on its next move and wins. The distinct empty completions lift by Lemma 5.2. This proves both rule labels `double_immediate_win` and `double_open_four`; the labels do not supply additional hypotheses. $\square$

Proposition 5.4 (next-turn finish and exhaustive critical completion). *Suppose that for every legal White move its child has no White five and at least one Black immediate winning point. Black wins within two actual moves. More generally, retain precisely the White children without a Black immediate winning point, verify their continuations, and require no White five in every child. Every omitted child is won by the same two-move argument.*

Proof. The chosen White child is exhaustive under one-move lifting. White has not won, and the Black completion witness in that child lifts to the actual board. Black then wins. The second statement applies this argument to each omitted child and uses the verified continuation for each retained one. These are `next_turn_finish` and `white_critical`. $\square$

Proposition 5.5 (the small-count open-four critical rule). *Suppose the parent has exactly three Black stones and two White stones. Enumerate all White children, retaining those that contain no template consisting of four consecutive cells with exactly three Black stones, one empty setup point, and both adjacent endpoints empty. Every omitted child is won within four actual moves from the parent.*

Proof. In an omitted child White has three stones. Black fills the setup point, giving four consecutive Black stones with two empty winning endpoints. On its next move White has at most four stones in total, so it cannot form five anywhere. White can occupy at most one of the two endpoints, and Black wins at the other. Counting the parent's White reply, Black setup, White reply, and Black finish gives four moves. The exact count $(|B|, |W|) = (3, 2)$ in `white_open_four_critical` is essential to this proof: a template alone in a position with more White stones does not exclude an intervening White win. $\square$

5.2 Counterthreat actions and a forced block

Let $\mathcal{A}_W$ contain, for every five-cell segment disjoint from B and containing exactly three old White stones, every empty point of that segment as an anchored move. Require that every unblocked segment containing at least three old White stones has exactly three; equivalently, the reconstruction rejects any White immediate winning point and any White five. The union of these actions is `threat_actions`.

Lemma 5.6 (all relevant White counterthreats). *Under these hypotheses White cannot win immediately. If a White move creates any White immediate winning point, that move belongs to $\mathcal{A}_W$.*

Proof. An immediate White win would require four old White stones in an unblocked five-cell segment, excluded by the hypothesis. A newly created White winning point has a five-cell witness containing four White stones after the move. It must contain the new stone, since the old position has no White winning point, and therefore contains three old White stones. The reconstruction enumerates its new point. The three old stones have span at most 4 in each axis, so the segment is a rigid witness and the enumerated anchored action covers its move in every preimage. $\square$

Proposition 5.7 (forced block). *If White has no immediate winning point and Black has an immediate winning point q , retain the anchored White action at q . Every other White reply loses within two moves.*

Proof. White cannot win now. A reply away from q leaves the Black completion at q available. A supporting Black stone adjacent to the hole exists in any four-of-five witness; using such a stone as reference gives the offset of norm at most 1 used by `white_forced_block_symbolic`. The completion and anchored block both lift exactly. $\square$

5.3 The fork rule

For each point s , collect targets T_s : a target $t \neq s$ is included when an unblocked five-cell segment contains exactly three old Black stones and has the two empty points s, t . Choose any s with at least two distinct targets. If $|T_s| = 2$, the fork blockers are $\{s\} \cup T_s$; if $|T_s| \geq 3$, the sole necessary blocker is s . Add these blockers to $\mathcal{A}_W$. The reference module chooses one such fork; optimality of that choice is irrelevant.

Lemma 5.8 (fork witnesses share a rigid frame). *All supporting Black triples of a reconstructed fork belong to one old rigid component, and all their displayed setup points represent the same point of that frame in every preimage.*

Proof. A supporting triple consists of three cells of a five-cell segment other than s . At most two cells of that segment lie at distance greater than 2 from s , so the triple has a stone at Chebyshev distance at most 2 from s . Choose one such stone from each triple. Any two chosen stones have

mutual distance at most 4, and are joined in the old-stone graph. Each triple is itself connected because its stones lie in one five-cell segment. Thus all triples lie in the same component. Their offsets to the common displayed s , and to each target, are exact within that component, proving the claim. $\square$

Proposition 5.9 (symbolic fork). *Every White move outside the reconstructed blockers and $\mathcal{A}_W$ loses within four moves from the parent.*

Proof. By Lemma 5.6 the reply neither wins immediately nor creates a White completion point. The setup s remains empty. If there are two targets, both survive because the reply avoided s and both targets. If there are at least three targets, a reply away from s can occupy at most one target and leaves at least two. Black plays s , obtaining two distinct completion points. Adding a Black stone cannot create a White winning point. White therefore cannot win on its next move, and Black completes at a surviving target. Lemma 5.8 establishes this construction in every preimage. This proves `white_fork_symbolic`, with the four-move budget including the initial White reply. $\square$

5.4 The open-four template rules

An open-four preparation template consists of a four-cell segment with exactly three Black stones and one empty setup s , and with both adjacent endpoints ℓ, r empty. Its **obstruction set** is $T = \{s, \ell, r\}$. A White reply disables this particular preparation only by occupying T , because its other required cells already contain Black stones. For a nonempty selected family of templates in a common rigid frame, let $K = \bigcap T$. Retain $\mathcal{A}_W$ and the anchored White moves at all points of K .

Proposition 5.10 (a selected template family). *Every White reply outside those retained actions loses within four moves.*

Proof. A reply outside K misses the obstruction set of at least one selected template. Its setup and both endpoints remain empty. By Lemma 5.6 the reply has no immediate or next-turn White win. Black fills that setup. It produces two Black completion points and can only reduce White's completion set. The next White move cannot win or block both endpoints; Black wins on the following move. Each template's support and endpoints lift within the selected rigid frame. $\square$

The four implementations select justified families as follows.

- `white_open_four_symbolic`: if there are at most four Black stones, select all templates. Each uses a three-element subset of B , so any two supporting triples share at least two Black stones and belong to one rigid component. If there are more than four Black stones, select a single template. In either case the selected frame is rigid. A common obstruction point belongs to the first template and is within distance 4 of any supporting stone chosen there as reference.
- `white_open_four_symbolic_shared`: require at least five Black stones and select a family whose templates all contain a chosen old Black supporter. Their offsets are exact from that common supporter. The common obstruction points lie within distance 4 of it.
- `white_open_four_symbolic_component`: require at least five Black stones, form the old-stone components including White connector stones, and select all templates supported in one component. Their offsets are exact by Lemma 5.1. Anchor common obstruction points at a supporter of the first selected template, so their offsets have norm at most 4.

- **white_open_four_symbolic_far**: require at least six Black stones and two templates with supporting stones u, v at displayed Chebyshev distance greater than 8. Each template's obstruction points are within distance 4 of every one of its supporters. If the two obstruction sets shared a point, the triangle inequality would give distance at most 8 between u, v , a contradiction. Coordinate expansion cannot reduce either axis separation between old points; the two sets remain disjoint in every preimage. Every White reply therefore leaves one preparation intact. Only $\mathcal{A}_W$ needs retention. The two templates can use separate rigid frames; this argument does not require their relative translation to be known.

All these rules use the four-move clock of Proposition 5.10. The chosen family and fork may be changed by one of the eight square symmetries. These symmetries preserve legal moves, winning directions, distances, templates, and budgets; applying and then undoing the chosen frame does not alter the proof.

6 Long threat sequences and remote interruptions

Throughout the local simulation, A denotes a simulated Black set and D denotes a virtual White upper set, not a set of directions. The winning directions remain those fixed in the board rules. All real boards are subsets of $\mathbb{Z}^2$, and normalized full-board successors use α .

For a Black-to-move full root after $p = |B| + |W|$ actual moves, let $h = H - p$ be its remaining budget. The cardinality of a virtual White upper set does not replace p : adding every defense of one operator counts as one actual White turn. Local subboards and virtual boards need not have equal Black and White cardinalities.

6.1 Local operators and their complete sufficient continuation

The following is an intrinsic description of the finite checks, independent of whether a program accepts them. Fix a rigid local frame. A step records a Black gain g , required Black support R , an empty set U containing g , a White defense set K , and one of the following line patterns. Coordinates in the table are consecutive indices along any winning direction; translating a pattern, reversing its line, and choosing any stone of its final Black shape as the gain are allowed. The remaining shape stones are the required support.

Pattern	Black shape after the gain	Defense set K
five	$\{0, 1, 2, 3, 4\}$	$\emptyset$
four	$\{0, 1, 2, 3, 4\} \setminus \{t\}$	$\{t\}$
straight_four	$\{1, 2, 3, 4\}$	$\{0, 5\}$
three or broken_three	$\{1, 2, 3, 4\} \setminus \{t\}$	$\{0, 5, t\}$
open_three	$\{2, 3, 4\}$	$\{1, 5\}$

Pattern	Required empty cells before the gain
five	$\{g\}$
four	$\{g, t\}$
straight_four	$\{g, 0, 5\}$
three or broken_three	$\{g, 0, 5, t\}$
open_three	$\{g, 0, 1, 5, 6\}$

In the fourth row the endpoint omissions $t = 1, 4$ receive the first label and interior omissions $t = 2, 3$ the second. These names have no logical force beyond their reconstructed pattern.

Lemma 6.1 (a reply outside a defense set). *After a legal nonterminal gain, assume White has no immediate winning point. An ensuing White reply outside K , which creates no White immediate winning point, loses within two moves for **four** and within four moves for one of the three-stone types. A **straight_four** is won within two moves against every White reply. A **five** ends immediately at the gain.*

Proof. For **four**, the reply misses the sole completion. For **straight_four**, it misses at least one of the two endpoints. For **three** and **broken_three**, Black fills t , giving four consecutive stones with endpoints 0 and 5 free. For **open_three**, Black can extend at 1 or 5. Extension at 1 requires endpoints 0 and 5, and extension at 5 requires endpoints 1 and 6. A reply outside $\{1, 5\}$ may occupy 0 or 6, but one stone cannot occupy both; choose the extension whose external endpoint remains free. In each three-stone case this next Black move makes a straight four. The preceding White reply created no White completion, and adding a Black stone cannot create one, so White cannot win on the intervening move. Black then finishes at a remaining endpoint. The counts include the White reply under discussion. $\square$

The finite local continuation maintains a state (i, A, D, u) , with step index i , simulated local Black set A , virtual White upper set D , and actual-move count u . The following conditions define a valid local witness.

1. Initially A, D are the old local stones and $u = 0$. At every step D has no five, the required support lies in A , and the required empty set avoids $A \cup D$. Add g to A and increase u by one. If Black has five, terminate this branch. Otherwise require $\text{Win}(D, A) = \emptyset$.
2. A **straight_four** terminates this branch with total bound $u + 2$. For the other nonterminal types form a closure of all counterfour responses described below, starting with the state just after the gain.
3. At every White-decision closure state (A, D, v) , require that D has no five and $\text{Win}(D, A) = \emptyset$. For a three-stone operator, enumerate every empty point q of every five-cell segment disjoint from A with at least three White stones in D . Put $D' = D \cup \{q\}$. The prestate conditions imply that D' has no five, since otherwise q was a completion of D . If $\text{Win}(D', A)$ is nonempty, require it to be the singleton $\{t\}$. Require q, t to avoid every gain, required support, defense point, and required empty point in the current and later operators. Add the full closure successor $(A \cup \{t\}, D', v + 2)$. No such response is ignored. A branch whose Black block already makes five may be treated as having terminated early.
4. At every closure state check $K \cap A = \emptyset$ and that $D \cup K$ has no five. Append the next step with virtual set $D \cup K$ and count $v + 1$. Also include the quick-finish bound $v + 4$ for a three-stone type, or $v + 2$ for **four**.
5. Every branch has a terminal gain or straight four before the sequence runs out. All closure states and all bounds are finite and within the stated local budget. A resource limit, missing step, extra completion point, collision, or excess budget invalidates the witness; a closure may not be truncated and then certified.

Call the maximum of these branch and quick-finish counts L . Let E contain all old local stones, every operator position, and every position added by any closure edge. Let $\mathcal{P}$ be the recorded local

states: the initial state, states after gains, states after counterfour blocks, and states after adding virtual defense sets. A local White counterfour state before its Black block is not in $\mathcal{P}$; both new move points nevertheless lie in E .

Lemma 6.2 (completeness of a local counterfour closure). *The enumeration covers every actual local counterfour on any sustained branch represented by $d \subseteq D$. An accepted virtual closure does not omit a response because a virtual defense already occupies the actual new point.*

Proof. At a White decision the accepted virtual state has neither a White five nor a White completion. Suppose an actual legal reply q creates a White completion t , witnessed by a five-cell segment S . Before q , the segment has three actual White stones and no actual Black stone, so its three-stone support is contained in D and the enumeration includes q , provided $q \notin D$.

If $q \in D$, then D already contains the resulting actual four-stone support. If $t \in D$, it already contains a five; otherwise t is a virtual White completion. Both contradict the accepted prestate. Thus $q \notin D$. Likewise an actual completion t cannot be hidden by virtual occupancy: if $t \in D$, the prestate has four stones of S , missing only q , and hence a virtual completion at q . Therefore the actual completion points persist among the virtual completion points after q . The virtual set must have exactly one completion point, so this point is also the unique actual completion and the prescribed Black block is correct. The virtual post-reply cannot already have five, because this would require a virtual completion in the prestate. The suffix exclusion preserves the current operator and every later recorded requirement. $\square$

Proposition 6.3 (local witness soundness). *On the local board a valid witness gives a Black strategy that wins before White within L actual moves.*

Proof. Maintain actual local Black stones equal to A , actual White stones contained in D , and the indicated actual clock. A prescribed Black gain is legal because its empty set avoids the stronger set $A \cup D$. If it makes five, the actual game ends. At each ensuing White decision an actual immediate win is excluded: its four old stones would give either a five or a completion in the virtual state.

If White occupies K , add all of K to the virtual set and count one actual White move. The next prescribed Black gain is legal under this stronger set; if that defense gave an actual completion threat, the checks after that gain ensure it is blocked before White moves again. If White instead makes a counterfour against a three-stone operator, Lemma 6.2 prescribes its unique block, counts two moves, and returns to the same operator. For **four** or **straight_four**, a White move that merely creates its own four loses to Black's already available five on the next move and does not require a counterfour detour. Every other reply is the quick-finish case of Lemma 6.1. These cases exhaust all legal White replies. A quick-finish branch ends and never returns to the sequence. The finite continuation conditions exclude unfinished branches; the specified maximum clock is L . $\square$

6.2 Envelopes and mixed-line guards

Fix an old rigid component C as local frame. For each old external point r , define its axis envelope as follows. If its coordinate lies between the extreme coordinates of C , or the axis path from it to the nearest extreme of C crosses no gap displayed as 5, its relative coordinate is exact. Otherwise a point to the left has envelope $(-\infty, r_a]$, and one to the right has envelope $[r_a, +\infty)$. Take the product of the two axis intervals, denoted $I_C(r)$, in the representative coordinates aligned with C .

Lemma 6.4 (envelope coverage). *Every relative location of r in every integer preimage lies in $I_C(r)$. If Q is another rigid component with old anchor r , a generated point x of its own frame has envelope $I_C(r) + (x - r)$.*

Proof. Inside the axis projection of C there is no stretchable gap by Lemma 5.1. Outside it, increasing any intervening saturated gap moves the point only farther in its existing axis direction; if no such gap intervenes its displacement is fixed. The rectangle drops dependencies between the axes and hence can only enlarge the true set. All points of Q move with its anchor plus their exact local offsets. $\square$

For a recorded local state (A, D) and a five-cell segment S meeting D but avoiding A , define

$$m_C(S) = \#\{r \in R : I_C(r) \cap (S \setminus D) \neq \emptyset\},$$

where R is the retained set of old remote White stones. Require

$$m_C(S) = 0 \quad \text{or} \quad |S \cap D| + m_C(S) < 3. \quad (\text{M})$$

This counts old stones, not lattice cells. The envelope events need not be simultaneously realizable, and different envelopes may hit the same point; counting them all is a safe overestimate. Remote Black stones are ignored in this guard, which also makes it stronger. Together with separation of envelopes from E , M implies that any truly unblocked line containing both local and remote White stones has at most two such stones at each recorded state. A real remote stone cannot fall on an already occupied virtual local point because those points lie in E .

For one remote White stone, it suffices to exclude its envelope from the union of all empty cells of all unblocked five-cell segments containing at least two members of D , for every recorded state. This is precisely M specialized to $|R| = 1$. A cache of this complete union changes no mathematical condition.

Lemma 6.5 (the unrecorded local intermediate state). *Testing M on $\mathcal{P}$, without storing the state immediately after a local White counterfour and before its Black block, is sufficient, provided the complete footprint and the following pure-region checks hold. This statement also applies to the multiple-frame guards in Subsection 6.4.*

Proof. Let q be the local White counterfour move from a recorded White-decision prestate, and t its prescribed block. If q also creates an immediate completion on a mixed line, that five-cell witness had three old White stones before q , all on an unblocked mixed line. The prestate guard forbids it. A White five on this move would similarly require four old stones and is excluded by that guard or by the pure-region checks. A completion arising purely in the local frame is included among all completion points in Lemma 6.2. A completion arising purely in another active frame makes the same q a counterfour point in that frame's complete graph, so q lies in both footprints, contradicting their separation. For a static remote set with every remote three blocked, a pure remote completion is impossible.

The new stone q might produce a mixed line with three stones but no immediate completion. Black now plays t ; no second White move occurs in between. If this new mixed line remains unblocked after t , it appears in the recorded post-block state and is rejected by its guard. If t blocks it, it cannot produce a later White threat, because Black stones persist. Thus the unrecorded intermediate cannot hide either an immediate extra completion or a usable future mixed three. $\square$

The local `positions` sets do not record the White counterfour intermediate. Their sufficiency is the precise statement of Lemma 6.5. The graph for a remote component, in contrast, explicitly records both intermediate and post-block states.

6.3 Infinite and prepared remote macros

Theorem 6.6 (infinite threat macro). *Let a nonterminal Black-to-move dynamic-gap state have all old Black stones in one old rigid component C . Let $A = B$, $D = W \cap C$, and $R = W \setminus C$. Assume a valid local witness of bound $L \leq h$, with its complete $E, \mathcal{P}$, and assume:*

1. *No five-cell segment in the representative contains three old members of R .*
2. *For every $r \in R$, its envelope avoids E .*
3. *Guard M holds for every recorded state and every indicated segment.*

*Then in every integer preimage Black has a strategy that wins first in at most L subsequent actual moves. This is the rule **infinite_threat_sequence**.*

Proof. Three remote stones on a five-cell segment in a preimage would have mutual axis span at most 4. Their displacements survive compression, giving the forbidden three in the representative. Thus no pure remote counterfour can arise by adding one stone, and there is no pure remote immediate win. The envelope guard ensures that no old remote White stone occupies a prescribed operator, counterfour, defense, or empty point. At each recorded decision, M excludes an unblocked mixed old three, and hence excludes a new mixed counterfour or White immediate five. Lemma 6.5 covers the omitted local counterfour intermediate states. Every sustained White response is therefore a local defense or a local counterfour and is simulated by Proposition 6.3. Every other response is beaten by Lemma 6.1 and ends the game. This includes arbitrary points outside every fixed finite neighborhood: their harmlessness follows from their lack of a current defense or completion, and the immediate finishing race. It does not require identifying their long-term normalized successors with a pass. The local clock remains $L \leq h$. $\square$

Theorem 6.7 (prepared remote macro). *Choose any old Black anchor and its old rigid component C . Put $A = B \cap C$, $D = W \cap C$, $P = B \setminus C$, and $R = W \setminus C$. Assume a valid local witness with $L \leq h$, and require:*

1. *Every representative five-cell segment containing at least three members of R contains an old Black stone of B .*
2. *Every envelope of every member of $P \cup R$ avoids the complete local footprint E .*
3. *Guard M holds throughout $\mathcal{P}$.*

*Then every integer preimage is won by Black first within L moves. This is **prepared_remote_threat_sequence**.*

Proof. A preimage remote triple on a five-cell segment has exact internal displacements. Its compressed triple lies on the corresponding representative segment, whose Black blocker is at distance at most 4 from each of those White stones. The blocker's displacement is exact too, so the blocker remains on the actual segment. This excludes every truly unblocked pure remote old triple. It also excludes any pure remote immediate four or five: such a segment contains a triple and cannot contain both a Black blocker and a White five. Existing blockers are never removed.

Old remote Black stones are retained in the actual game. They may block additional White lines but cannot occupy any prescribed local point by condition 2. Ignoring them while testing local and mixed White threats is conservative. Thus all operator requirements and blocks lift, and all pure remote or mixed threats excluded in Theorem 6.6 remain excluded. Apply its reply classification and Proposition 6.3. An actual Black block that forms five with remote Black stones ends the game earlier. No clock is added for merely retaining those old stones. $\square$

6.4 Complete remote interruption graphs and arbitrary interleavings

Partition all old stones outside the local component C into their distinct old rigid components $Q_1, \dots, Q_k$. Each component has its own anchor and exact offsets. Treating their union as a single rigid region would be unsound.

For each Q_j , form a finite complete graph of White counterfour moves with automatic Black blocks. Initially take its old Black and White stones, assume neither color has five and White has no immediate completion. At a graph state (A_j, D_j) with no Black five, enumerate every empty point of every five-cell segment avoiding A_j and containing at least three D_j stones. Retain every point q for which $D_j \cup \{q\}$ has any completion. Require that it has no five and exactly one completion t . Its edge is

$$(A_j, D_j) \longrightarrow (A_j \cup \{t\}, D_j \cup \{q\}).$$

Both the intermediate White position and the final position must be recorded. A Black-five state terminates its path. Expand every other successor completely. Let N_j be the maximum number of edges on any graph path, F_j its full old and generated footprint, and $\mathcal{P}_j$ its full position set. No boundary, depth cutoff, or timeout may count as a completed graph. Finiteness and the claimed maximum are hypotheses verified by exhaustive closure, not inferences from a sample path.

Choose a valid local witness with bound L . Translate each F_j and each state of $\mathcal{P}_j$ by its rigid anchor envelope as in Lemma 6.4. Require:

- Every remote footprint envelope avoids E . In each remote reference frame every other remote footprint envelope avoids F_j . This is pairwise separation of all future activity, not merely separation of old stones.
- For every $(A, D) \in \mathcal{P}$ and every five-cell segment S meeting D and avoiding A , define $m_j(S)$ as the maximum, over $\mathcal{P}_j$, of the number of that state's White stones whose translated envelopes can meet $S \setminus D$. Require

$$\sum_j m_j(S) = 0 \quad \text{or} \quad |S \cap D| + \sum_j m_j(S) < 3. \quad (\text{MI})$$

- In each Q_i frame apply the same guard to every recorded state in $\mathcal{P}_i$, with all other remote components supplying their statewise maximum contributions. This checks mixed lines that contain no local White stone. Remote Black blockers can be ignored in these guards.
- The actual clock satisfies $L + 2 \sum_j N_j \leq h$.

The maxima and their sum may combine states, positions, or translations that can never occur together. This can reject a usable witness, but covers every attainable product state and every possible ordering of graph edges.

Theorem 6.8 (remote interruption macro). *Under these hypotheses Black wins first, against every legal future White play in every integer preimage, within*

$$L + 2 \sum_{j=1}^k N_j$$

actual moves. This is `remote_interruption_threat_sequence`. A single remote component is the case $k = 1$.

Proof. Fix an arbitrary preimage and realize each rigid frame by its actual old anchor. Follow the local strategy of Proposition 6.3. On a sustained branch maintain:

1. Actual local White stones are contained in the current virtual local D , and actual local Black stones are the current local A . Each remote region is at one state of its complete graph; any completed Black-five graph state has already ended the actual game.
2. A sustained White response is a local defense, a local counterfour, or a counterfour in one remote frame. A response outside these cases invokes the quick finish and does not re-enter the simulation, so sustained branches do not accumulate unlisted quiet White stones or new regions.
3. At each White decision there is no unhandled White completion. Pure local completions are excluded by the local checks and Lemma 6.2. Pure remote completions are excluded initially and after each unique block. Any mixed completion, direct White win, or new mixed counterfour would require an unblocked old mixed three (indeed four for a direct win), excluded by the mixed guards.
4. Every prescribed point is legal and affects only its assigned frame, because all activity footprints are separated in every preimage. The local clock is at most L , and the number of edges used in region j is at most N_j .

Here “assigned frame” is determined by the old three-stone support of the threatening five-cell line, rather than by the displayed location of the new point. The guards prohibit supports split across two or more frames. For mixed lines containing a local White stone, MI covers all remote states. For lines without any local White stone, choose a remote frame containing a White stone of the line and use its remote-frame guard. Hence lines mixing three or more remote components are also covered. Virtual local points cannot hide an actual local counterfour by Lemma 6.2, and footprint separation excludes their occupancy by remote stones.

After a local gain, White’s present move cannot make five. If it is a local defense, simulate it by the virtual defense union and continue. If it makes a local counterfour at a three-stone operator, use the local closure block. Lemma 6.5 ensures that its omitted intermediate state cannot conceal an additional mixed or pure remote completion, or leave an untested mixed three. If it makes a counterfour in Q_j , its graph gives the unique completion t ; Black plays t . The graph records the intermediate and post-block state, and its edge has consumed two actual moves. Neither point affects the local operator or another graph. Thus the same local White decision is reached again with only the Q_j state advanced. If one move created pure counterfours in two frames, its new point would lie in both complete activity footprints, contradicting separation. A fork within one remote frame would give at least two completions and invalidate its graph.

For a local four, any White reply not preventing Black’s next five loses on that next Black move, even if it creates a remote four. For a local three, a reply which neither defends nor makes any completion is the quick-finish case of Lemma 6.1. Its lack of a White completion is a statement about the full actual board; therefore the next Black extension cannot create a White winning point, and White cannot win on the intervening move. This argument applies also to previously unlisted points at arbitrary distance.

These cases exhaust White’s legal play and preserve the four invariants on every sustained branch. Local and remote counterfour pairs may alternate in any order. The statewise maxima in the mixed guards cover their full product, and pairwise footprint separation preserves their individual legal moves. Every remote edge adds one actual White stone and one actual Black stone;

region j can use at most N_j such edges over the entire play. Therefore all remote delays together cost at most $2 \sum_j N_j$, while the local progress, local counterfours, defenses and quick finish cost at most L . Finiteness gives termination and the required bound. $\square$

The explicit two-component example and strategy dispatcher in Appendix F show the four reply cases, the recorded states, and the clock accounting in a small fully checked certificate.

7 The opening cover and square symmetries

After Black plays the origin, an actual White reply $q \neq (0,0)$ normalizes by independently truncating its signed coordinate magnitudes to five. The normalized reply set is

$$R = \{-5, -4, \dots, 5\}^2 \setminus \{(0,0)\}, \quad |R| = 120.$$

Here $\alpha(q)$ abbreviates the White coordinate in the normalization of $B = \{(0,0)\}$, $W = \{q\}$; it does not denote a physical move of q . The square symmetry group D_4 consists of the eight signed coordinate permutations. A convenient representative set is

$$T = \{(-a, -b) : 1 \leq a \leq 5, 0 \leq b \leq a\}. \quad (2)$$

Every orbit in R has exactly one representative in T , and $|T| = \sum_{a=1}^5 (a+1) = 20$. Its size is four when $b = 0$ or $b = a$ and eight otherwise. These sizes sum to 120.

Lemma 7.1 (Normalization commutes with D_4). *For every signed coordinate permutation M and finite colored position, $\alpha(MP) = M\alpha(P)$.*

Proof. Swapping axes swaps their independent normalization maps. Reversing the sign of an axis reverses the ordering of its values and preserves the list of adjacent gap lengths in reverse order. Since zero is the anchor, the resulting map is exactly the negative of the original map. $\square$

Proposition 7.2 (Opening lifting). *Suppose for every $t \in T$ there is a certificate for Black to move after Black at $(0,0)$ and White at t , sound for every concrete preimage and with the same total bound H . Then Black wins from the empty board within H total placements.*

Proof. Black starts at the origin. For the actual legal White point q , find the representative t and a signed permutation M with $Mt = \alpha(q)$. Apply M^{-1} to the actual board, not merely its drawing. Lemma 7.1 gives $\alpha(M^{-1}q) = t$. The representative strategy therefore applies to that actual preimage. Transport every prescribed Black placement back by M . A signed permutation preserves emptiness, the four winning directions, and the order and number of actual moves. The representative root has already consumed two placements, so its bound H is counted from the empty board, not restarted after White's reply. White is allowed any subsequent legal point, as already required by the representative certificate. $\square$

An actual distant coordinate need not equal its representative. For example, $(-10^6, -10^{30})$ belongs to the representative $(-5, -5)$ but the actual stones remain at their actual locations. The strategy uses the relative action semantics and the complete uncertainty successors of Section 3. It does not physically replace the distant White stone by a nearer one.

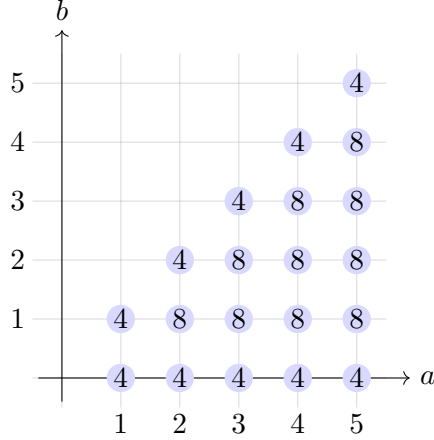


Figure 3: The 20 representatives $(-a, -b)$. Numbers are orbit sizes, not search counts. The 10 boundary representatives have size four and the 10 interior representatives have size eight, covering all 120 normalized first replies. A coordinate magnitude five also covers all larger actual magnitudes of that coordinate.

8 Independent verification and observed computation

8.1 What the checker trusts

The independent checker is an eight-source Python stack using the standard library. It does not import the search implementation, the first checker, RV2, or Rapfi. It reconstructs normalized axis insertions and five-segments from finite sets; reconstructs the critical replies of symbolic rules; checks the sequence sufficient conditions; and follows the complete modular proof closure. At the opening layer it reconstructs R , checks all 20 roots and all 120 representative assignments, validates each symmetry matrix, and requires an empty-board interpretation. Assertions must remain enabled; execution with Python’s `-O` option is rejected. A saved receipt is evidence of an earlier computation, not a substitute for a fresh mathematical check in the final run.

The frozen local identities are listed below. Digests identify records; they do not supply the complete certificate or replace fresh verification.

Empty cover	fce6b466f2675cb5a76f40b1e979fb77bdfc4abc6971971510e1c60196adf85c
Mathematical result	9720a055b970aa092dc216ca73059055d97d040ae3c47a31f357b26a7b40443b
Byte audit	226562fd83b790dbce3f076aedad58e7ff22f9ec4d8da78fe29982cd448b90f4

8.2 Completed run and scope of the statistics

The final mathematical run checked all 20 opening roots and all 120 normalized first replies, with total bound 35 and no reused checked-file receipts. The observed process exited with status zero. The 20 roots used sequential four-worker pools, with every observed worker exiting successfully. The fresh mathematical entry point reported 27,837.450 seconds. The enclosing wrapper reported 27,837.498 seconds; the latter contains the former and they must not be added.

A separate complete byte audit read the selected dependency closure and checked it against the fresh receipts. It reported 1,482,906 distinct dependency paths and 1,652,612 file occurrences across roots, in 5,729.200 seconds. Distinct paths are not necessarily distinct contents. The audit’s actual process also exited with status zero. An operating-system cache prefetch helper ran during that audit; each digest was still recomputed from actual file bytes. Therefore this duration is not a cold-cache benchmark.

Table 1 records occurrences summed across opening roots. Some subproofs are shared, so those sums differ from the distinct path union. The node total is not the number of moves in a game or the number of distinct infinite-board positions. These are observations of a particular verified proof object, not lower bounds on certificate size.

Quantity	Observed value
Normalized first replies covered	120
Symmetry representative roots checked	20
Ordinary proof-file occurrences	1,644,613
Ordinary reachable-node occurrences	109,364,098
Bundle-file occurrences	7,999
Bundle-edge occurrences	1,954,756
Distinct selected dependency paths	1,482,906
Distinct-path reachable nodes	102,044,378
Selected JSON payload, bytes	48,588,463,288
Fresh mathematical check, wall seconds	27,837.450
Separate full byte audit, wall seconds	5,729.200

Table 1: Recorded certificate and verification statistics. Occurrences can repeat across roots. Mathematical phases, wrapper time, and audit time are not independent CPU-cost measurements. No complete from-zero search time or overall search speedup is inferred from these numbers.

The manuscript-time environment was an Intel Core i7-12700H, 20 logical processors, Windows 11 Pro build 26200, about 31.7 GiB memory and system Python 3.13.14. These are not historical start-time records. Each replication must record its actual runtime, hardware, exit status and wall time.

8.3 Rule correspondence and version binding

Appendix E gives the completed census: 12 of the 17 ordinary rule labels occur; the other five have zero uses. Prepared remote sequences occur 971,354 times and remote interruption sequences 64,888 times. The census read every selected dependency and checked its digest against the sealed fresh receipts; it counts reachable labels, not mathematical replay. Its CPython 3.12.14 Windows run reported 25,255.734 seconds. Full counts and bindings are supplied in `rule-use-counts.csv` and `certificate-census.json`.

The accepted run binds the eight checker sources, actual roots, and selected proof bytes. A nonzero exit, missing dependency, stale receipt, source or byte mismatch, incomplete coverage, or unfinished rule closure is a failure, never a proof. Appendix D records the earlier failed runs and the final record-integrity gate.

Proof of Theorem 1.1. The completed independent run accepted the 20 representative-root certificates, each with the specified rules, actual two-stone root, and total-placement bound 35, and checked the 120 first-reply assignments. The accepted short and long rules have the uniform strategy properties proved in Sections 5 and 6. By Theorem 4.1, each representative certificate wins from every actual preimage within its remaining actual clock. Proposition 7.2 then transports these strategies to every actual legal first reply after Black starts at $(0,0)$. The 35 count includes those first two placements. The byte audit binds this calculation to the selected actual proof bytes; its

record is an integrity witness for the finite computation, not an additional mathematical assumption about distant play. $\square$

9 Search as an untrusted producer

The producer uses finite-board strategies, RV2, Rapfi, freestyle threat generation and geometric ordering as Black action suggestions. Exact-five rules and finite engine windows can give inappropriate advice; only the infinite-board certificate checker accepts a strategy.

Search is a persistent AND/OR computation. Black chooses one action, but must prove all its abstract outcomes; White requires all abstract replies. Alternative actions keep separate obligations. Completed proofs and spent budgets persist across recovery. UNKNOWN, timeout, exhausted candidates and missing branches prove neither a White win nor a Black loss. A counterwin rejects the tested Black proposal, not every strategy. The chosen bound 35 is established for this certificate and is not a conjectured upper limit for all possible winning strategies.

10 Reproducibility and review materials

The [complete review release](#) supplies the empty cover, all 1,482,906 selected dependency files, frozen verification sources, file manifests, licenses, and recorded successful verification. The [source repository](#) also provides the manuscripts, rule census, archive tools and examples. The archive preserves the original certificate bytes and relative paths, so the cover and checker identities in Section 8 remain valid. Selected JSON dependencies total 48,588,463,288 bytes (45.25 GiB), excluding meta-data and verification receipts. Exact compressed volume sizes and SHA-256 values are supplied in `archive-manifest.json`.

Download all numbered volumes and the release’s extraction entry, which checks their hashes and reconstructs the directory tree. Verification uses 64-bit Windows and Python 3.12 or newer, with only the standard library. From the extracted package root, run this one copyable line:

```
python -B verify-full-review.py --workers 4 --output-dir fresh-check
```

The output directory must be new and assertions enabled. The entry checks shipped bytes and source identities, calls the frozen mathematical checker on the entire closure, and seals the inputs again. Successful output is actual exit status zero and `fresh-check/release-acceptance.json` with status `VERIFIED_EMPTY_BOARD_REVIEW_RELEASE`, 20 accepted roots, 120 covered first replies, and total-placement bound 35. Errors, altered bytes, missing dependencies or unfinished coverage are rejected. Historical receipts document the recorded run; they are not premises of this fresh check.

The recorded mathematical run took 27,837.450 wall seconds with four workers on the machine described in Section 8. Allow storage for the extracted JSON corpus, downloaded volumes and new receipts; the release README gives the extraction and verification steps. Verification requires no search engine or search cache. Code is licensed under MIT; manuscripts and certificate data under CC BY 4.0.

11 Discussion

Distant play is covered by complete insertion sets and checked sufficient rules. A remote threat can force real defensive moves; its cost is included in the long-rule clock. Failure of isolation or counterthreat conditions requires explicit expansion. Conservative abstraction can enlarge proof

trees, so refined rules and alternative Black actions may reduce size, but need their own soundness and verification arguments. Neither optimality of 35 nor a smaller universal remote-play bound is claimed.

A Opening-root statistics

The companion `opening-run-statistics.csv` gives all 20 root observations. Ordinary file and node counts are occurrences within each root; bundle files are separate. Per-root seconds include header, mathematical-check and binding phases and are already included in the global wrapper time. The coverage argument is in Section 7, independently of these diagnostics.

B Checker sources and byte identities

The following eight files are one independent checker stack, not eight independent implementations of the whole theorem: All paths are in `scripts/` and have prefix `reference_`.

<code>gap_opening_cover.py</code>	<code>gap_bundle_batched.py</code>
<code>gap_bundle_parallel.py</code>	<code>gap_bundle_verifier.py</code>
<code>gap_verifier.py</code>	<code>infinite_threat_witness.py</code>
<code>prepared_remote_sequence.py</code>	<code>remote_interruption_sequence.py</code>

The publication inventory records their complete digests, the actual proof roots, and the completed mathematical and byte-audit records. Digest checks are necessary for identifying the verified version; they do not authenticate an observer or establish mathematical correctness by themselves.

C Contributions and research tools

The author provided the research objective, strategic guidance, and project supervision. OpenAI Codex with language-model assistance was used for research exploration, programming, debugging, and manuscript preparation; exact model identifiers are not established for every historical interaction. RV2 and Rapfi supplied untrusted move proposals. Their evaluations are not acceptance criteria, and the independent checker imports neither the search implementation nor either engine.

D Computational provenance and relocation

Earlier runs failed with a broken worker pool, audit exit status 120, or Windows error 5 during progress-file replacement. None was accepted or included in the successful timings. The final audit added only bounded replacement retries; mathematical checks and hashing were unchanged, and persistent errors still failed. Its completion gate binds successful runs, roots, source identities and bytes; it is not another mathematical traversal. Later dashboard edits did not change the frozen eight-source kernel.

The release retains the original proof bytes and Windows relative reference paths. The shipping manifest enumerates the complete selected closure; the review entry checks that every declared external reference is bound to a file within that closure. The mathematical checker reconstructs all retained successor obligations and validates their roots and digests. Archive-volume checksums identify the delivered bytes. The original recorded verification and final byte seal are supplied as computational provenance.

E Rule use and checker correspondence

Table 2: Rule-to-checker correspondence and final-certificate use. Counts are root-reachable ordinary-node occurrences summed over the 20 opening roots, not distinct positions or execution paths.

Node kind	Result	Entry	Checked premises; additional actual moves	Uses
terminal	4.1	G.v	Black just made five; White has no five; 0.	16,990
black	3.4	G.v	Legal anchored action, all its successors; 1 per edge.	33,273,731
white	3.3	G.v	Every normalized White successor; 1 per edge.	134,032
double_immediate_win	5.3	G.v	No White completion; at least two Black completions; 2.	11,042,442
double_open_four	5.3	G.v	The same premises as double completion; 2.	0
next_turn_finish	5.4	G.v	Every White child has no White five and a Black completion; 2.	0
white_critical	5.4	G.v	No White child wins; retain all without Black completion; omitted replies: 2.	0
white_open_four_critical	5.5	G.v	Exactly $(B , W) = (3, 2)$; retain all children without a template; omitted replies: 4.	0
white_forced_block_symbolic	5.7	G.c	No White completion; Black completion; retain its block; omitted replies: 2.	13,998,016
white_fork_symbolic	5.9	G.c	Rigid fork, all fork blockers and White counterfours; omitted replies: 4.	11,176
white_open_four_symbolic	5.10	G.c	Rigid selected template family and all counterfours; omitted replies: 4.	230,240
white_open_four_symbolic_shared	5.10	G.c	At least five Black stones; templates share an old Black supporter; omitted replies: 4.	0
white_open_four_symbolic_component	5.10	G.c	At least five Black stones; supports in one old rigid component; omitted replies: 4.	9,229,760
white_open_four_symbolic_far	5.10	G.c	At least six Black stones; two templates have supporters farther than 8; omitted replies: 4.	3
infinite_threat_sequence	6.6	I.v	All B in C ; local closure; no remote White triple; separated envelopes; M ; $L \leq H - p$.	40,391,466
prepared_remote_threat_sequence	6.7	P.v	Local closure; remote triples blocked; all remote stones avoid E ; M ; $L \leq H - p$.	971,354
remote_interruption_threat_sequence	6.8	R.v	Complete local/remote closures; unique blocks; separated envelopes; all-state MI ; $L + 2 \sum_j N_j \leq H - p$.	64,888

G, I, P, R denote respectively `reference_gap_verifier.py`, `reference_infinite_threat_witness.py`, `reference_prepared_remote_sequence.py`, and `reference_remote_interruption_sequence.py`, all in `scripts/`; v means `verify`, c means `critical_actions`. Except for terminal nodes, the parent is nonterminal, the turn and actual placement count are checked, and all retained actions require every normalized successor. The template variants retain the complete White counterfour set. Modular Black/White headers use `reference_gap_bundle_verifier.verify_file` with the same explicit successor rules; their counts are reported separately.

F A complete two-component example

The illustrative certificate `two-remote-components-example.json` has the following already normalized root, with 18 actual placements and Black to move. It illustrates the rule, not new opening coverage.

Frame	Old Black points	Old White points
C	$(0, 0), (1, -1), (1, 0), (3, 4), (4, 3)$	$(1, 3), (2, 3), (3, 3)$
Q_1	$(-8, 1), (-8, 2)$	$(-7, 0), (-6, -1), (-5, -2)$
Q_2	$(8, 10), (8, 11)$	$(9, 9), (10, 8), (11, 7)$

Local gains are $(1, -2), (2, -2), (0, -2), (2, 0), (-1, 0)$: four open threes followed by a straight four. Their defense sets, in order, are

$$\{(1, -3), (1, 1)\}, \quad \{(3, -3), (-1, 1)\}, \\ \{(-1, -2), (3, -2)\}, \quad \{(-1, -3), (3, 1)\}, \quad \{(-2, 0), (3, 0)\}.$$

The complete local closure has $L = 13$, footprint size 33, and 28 recorded states: initial, post-gain, post-block, and post-virtual-defense states. Local counterfour intermediates are handled by Lemma 6.5.

The local counterfour points $(0, 3), (-1, 3)$ are blocked at the other point. The remote alternatives are likewise $(-4, -3), (-3, -4)$ in Q_1 and $(12, 6), (13, 5)$ in Q_2 . Each remote graph records its initial state, two White intermediate states, and two Black post-block states; neither post-block state has a further edge. Thus $N_1 = N_2 = 1$. Both checkers reconstruct all graphs, envelopes and mixed guards, giving $L + 2(N_1 + N_2) = 17$ further placements and total bound 35. The supplied recorded-state file contains the complete state sets.

Strategy dispatcher. Play each prescribed local gain, stopping at a Black five. At its White decision, observe the actual reply and apply these cases in order:

1. If a four or straight four retains a Black completion after the reply, play it and stop.
2. If the reply is in K , add all K to virtual D , charge one local move, and advance to the next gain.
3. Otherwise, if a three-operator reply creates a White completion, locate its pure-frame support. Follow the complete local closure or the appropriate remote graph, play its unique block, and retain the operator. Charge two local moves or two moves to that remote graph; stop if the block makes Black five.
4. Otherwise use the operator's quick finish and stop.

The checked guards exclude a White win at each decision. Local charges are at most L ; remote charges are at most $2\sum_j N_j$ in every ordering. For example, after the first gain a quiet reply at $(1000, 1000)$ permits extension at $(1, 1)$ and then completion at $(1, -3)$ or $(1, 2)$.

The supplied recorded-state file also lists a legal continuation with both remote interruptions, a local counterfour, and ordinary defenses, ending with a Black win at total placement 35. This path illustrates the complete certificate; it is not a replacement for checking all branches.

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